

# Jetson Orin IMX415 Camera Driver Tutorial

---

## Jetson Orin IMX415 Camera Driver Tutorial

1. Scope of Application
2. Repository File Description
3. Pre-installation Confirmation
4. Copy Files to Orin
  - 4.1 Copy from PC via SCP
  - 4.2 Confirm the Files Have Arrived on the Orin
5. Back Up the Original Files
6. Install the Device Tree and Driver
7. Check the extlinux Configuration
  - 7.1 Direct Overwrite Configuration
  - 7.2 jetson-io.py Configuration
8. Reboot
9. Check Whether the Driver Is Loaded
10. View Kernel Logs
11. Check the Camera Device Node
12. Test the Image Stream
13. FAQ
  - 13.1 `nv_imx415` Is Not Loaded
  - 13.2 Driver Loaded but No `/dev/video*`
  - 13.3 dmesg Permission Issue
  - 13.4 `nv_imx415` Shows `Used by: 0`
  - 13.5 Overlay Not Loaded After Boot
  - 13.6 Switch Back to the Original Configuration
14. Notes

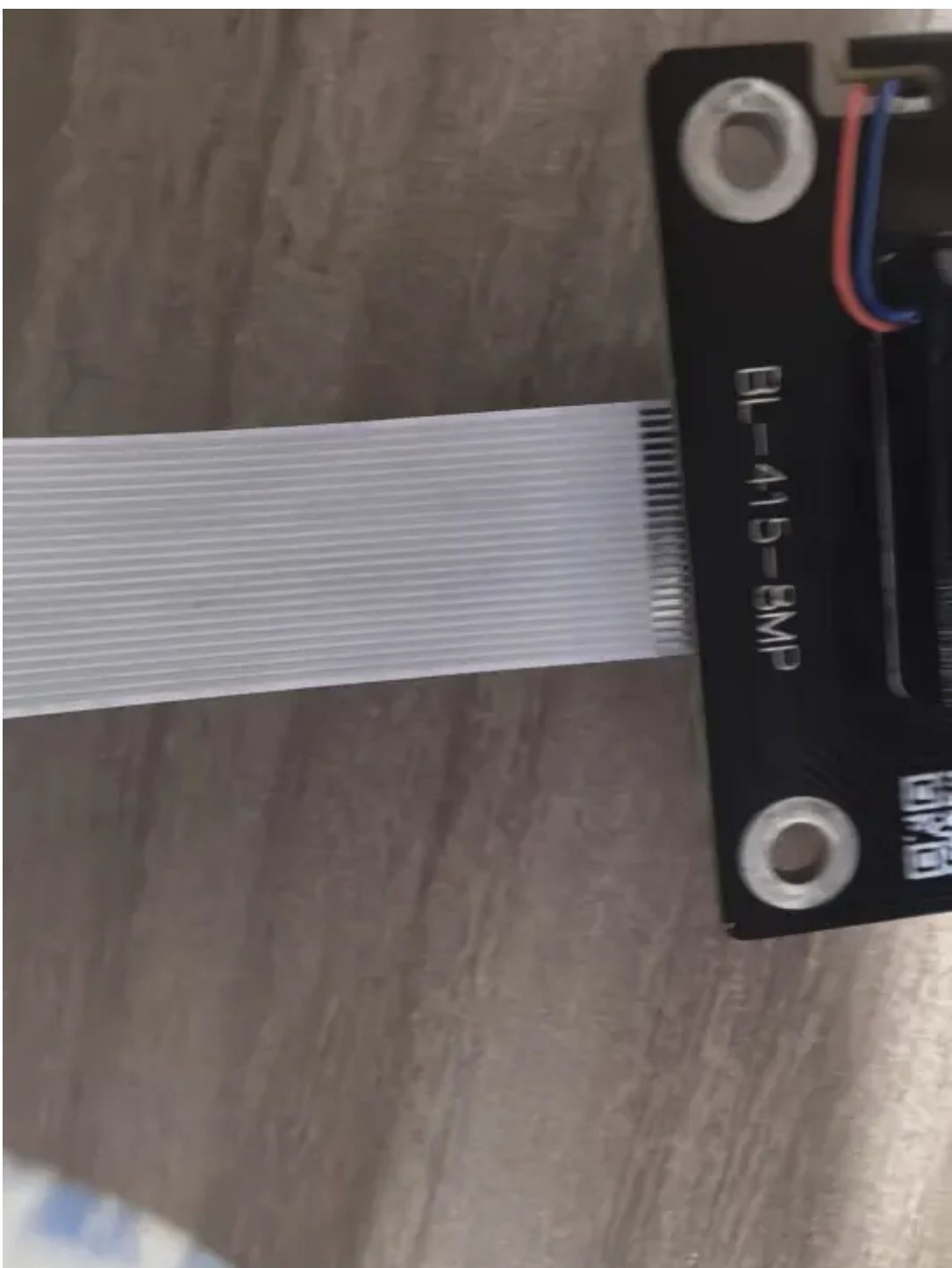
**Tip: If you need to install a bracket, insert all 4 screws at the same time first, then tighten them.**

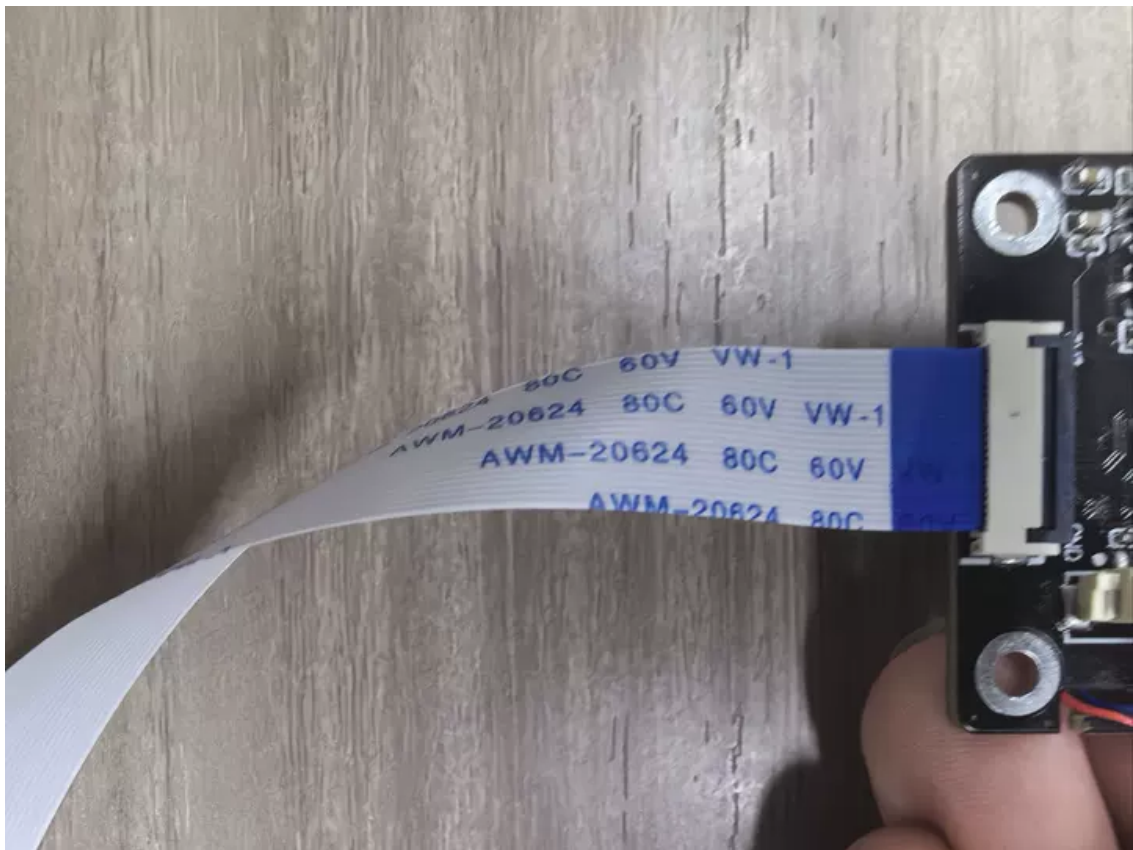
### Notes:

- **Contact surface position:** The metal contacts at the end of the ribbon cable must face the spring contacts or contact array inside the FFC connector.
- **Blue reinforcement board:** The blue reinforcement board is usually located on the non-contact surface of the ribbon cable. In standard installation, the metal contacts and the blue board are usually on opposite sides of the ribbon cable.
- **Silkscreen marking:** The side of the ribbon cable with silkscreen text (such as AWM-20624) is usually the back. During installation, it should face away from the contact surface (the gold or silver metal part).

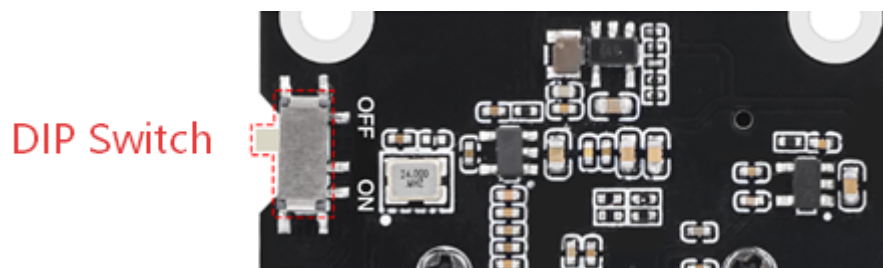
The metal must face toward the PCB side, otherwise the ribbon cable may burn out.

Then connect the other end to the metal contacts of the motherboard.





**Note:** The IMX415 has a DIP switch on its side called IR-CUT, which controls the infrared function. When set to the ON position, infrared is enabled and the image appears reddish — this is normal. When set to the OFF position, infrared is disabled and the image shows normal colors.



## 1. Scope of Application

---

This tutorial applies to the Jetson Orin environment corresponding to the repository:

- Kernel version: `5.15.148-tegra`.
- Boot method: `extlinux`.
- Carrier board/device tree: Based on `kernel_tegra234-p3768-0000+p3767-0005-nv-super-modify.dtb` and the HDR40 configuration in the repository.
- Camera: IMX415, 4-lane MIPI CSI.

Do not directly apply the precompiled files to other kernel versions or other carrier boards. The `vermagic` of `nv_imx415.ko` must be `5.15.148-tegra`.

**If you are using our latest factory image, no action is required. If you are using an old image or another image and want to use the IMX415 camera, follow this guide.**

## 2. Repository File Description

---

Files required:

```
jetson_orin/
├─ boot/
│   └─ extlinux/
│       └─ extlinux.conf
│       └─ tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo
└─ lib/
    └─ modules/
        └─ 5.15.148-tegra/
            └─ updates/
                └─ drivers/
                    └─ media/
                        └─ i2c/
                            └─ nv_imx415.ko
```

Corresponding installation locations:

| File                                            | Target path on Orin                  |
|-------------------------------------------------|--------------------------------------|
| tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo | /boot/                               |
| extlinux.conf                                   | /boot/extlinux/                      |
| nv_imx415.ko                                    | /lib/modules/5.15.148-tegra/updates/ |

### 3. Pre-installation Confirmation

Run on the Orin terminal:

```
uname -r
```

Must output:

```
5.15.148-tegra
```

Confirm that the original boot configuration and base device tree exist:

```
ls -l /boot/extlinux/extlinux.conf
ls -l /boot/dtb/kernel_tegra234-p3768-0000+p3767-0005-nv-super-modify.dtb
```

It is normal if this kernel\_tegra234-p3768-0000+p3767-0005-nv-super-modify.dtb file is different; you should operate based on your actual files.

Confirm that the camera is connected to the Orin's 4-lane MIPI CSI interface, and that the FPC ribbon cable direction is correct and locked firmly.

There are 2 ways to configure this. For details, refer to the files in the corresponding downloaded folder. The operations are roughly the same, but they differ in the final configuration step.

-  [Direct Overwrite Method](#) 2026/9/2 18:00
-  [Jetson-io.py Configuration Method](#) 2026/9/2 18:00

## 4. Copy Files to Orin

### 4.1 Copy from PC via SCP

Assuming the Orin's IP is `192.168.11.99` and the user is `jetson`:

```
scp jetson_orin/boot/tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo
jetson@192.168.11.99:/home/jetson/Desktop/

scp jetson_orin/boot/extlinux/extlinux.conf
jetson@192.168.11.99:/home/jetson/Desktop/

scp jetson_orin/lib/modules/5.15.148-tegra/updates/drivers/media/i2c/nv_imx415.ko
jetson@192.168.11.99:/home/jetson/Desktop/
```

**You can refer to the file transfer tutorial to transfer files. The purpose of this step is mainly to transfer the files to your board for easy operation. Transfer them to the corresponding directories, and adjust the commands in the following steps based on your actual paths.**

### 4.2 Confirm the Files Have Arrived on the Orin

```
ls -l /home/jetson/Desktop/tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo
ls -l /home/jetson/Desktop/extlinux.conf
ls -l /home/jetson/Desktop/nv_imx415.ko
```

## 5. Back Up the Original Files

Back up before modifying the boot configuration:

```
sudo cp /boot/extlinux/extlinux.conf /boot/extlinux/extlinux.conf.bak
```

If the target file already exists, back it up first as well:

```
if [ -f /boot/tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo ]; then
    sudo cp /boot/tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo
    /boot/tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo.bak
fi

if [ -f /lib/modules/5.15.148-tegra/updates/drivers/media/i2c/nv_imx415.ko ];
then
    sudo cp /lib/modules/5.15.148-tegra/updates/drivers/media/i2c/nv_imx415.ko
    /lib/modules/5.15.148-tegra/updates/drivers/media/i2c/nv_imx415.ko.bak
fi
```

## 6. Install the Device Tree and Driver

```
sudo cp /home/jetson/Desktop/tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo
/boot/tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo

sudo cp /home/jetson/Desktop/extlinux.conf /boot/extlinux/extlinux.conf

sudo cp /home/jetson/Desktop/nv_imx415.ko /lib/modules/5.15.148-
tegra/updates/drivers/media/i2c/nv_imx415.ko

sudo depmod -a

sudo modprobe nv_imx415
```

## 7. Check the extlinux Configuration

### 7.1 Direct Overwrite Configuration

View the configuration file:

```
cat /boot/extlinux/extlinux.conf
```

Focus on confirming:

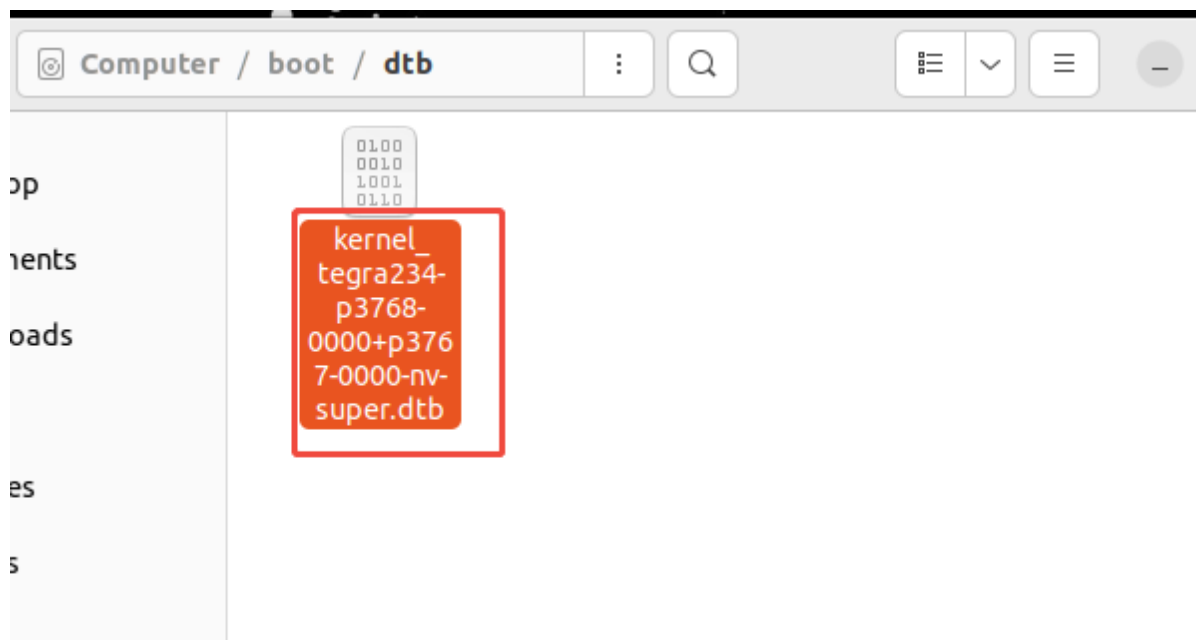
```
DEFAULT JetsonIO
```

And that the `JetsonIO` section contains:

```
FDT /boot/dtb/kernel_tegra234-p3768-0000+p3767-0005-nv-super-modify.dtb
OVERLAYS /boot/tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo
```

The `OVERLAYS` line must not be commented out, must not be another overlay, and the path must match the file under `/boot/`.

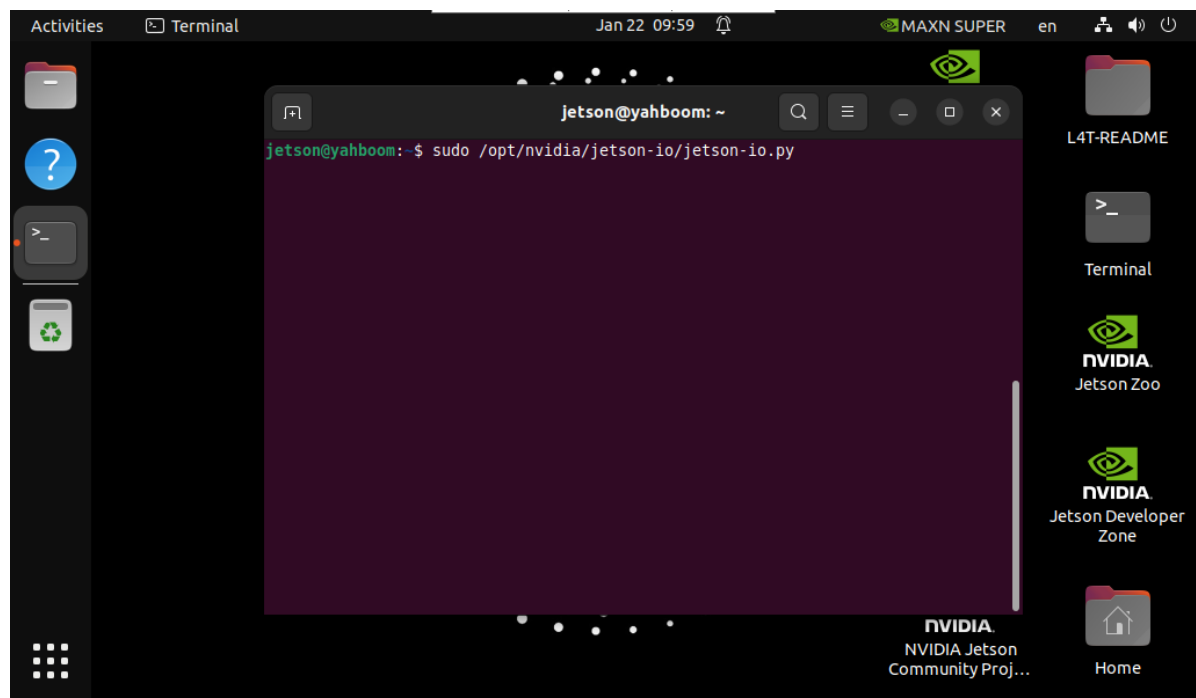
**The `kernel_tegra234-p3768-0000+p3767-0005-nv-super-modify.dtb` file must be consistent with the file in the boot folder. For example, if the file on the Orin board in the picture is called `kernel_tegra234-p3768-0000+p3767-0000-nv-super.dtb`, you need to change your FDT line to `FDT /boot/dtb/kernel_tegra234-p3768-0000+p3767-0000-nv-super.dtb`, otherwise the device tree configuration file cannot be loaded. In severe cases, the system may even fail to boot.**



## 7.2 jetson-io.py Configuration

Enter the following command in the terminal:

```
sudo /opt/nvidia/jetson-io/jetson-io.py
```



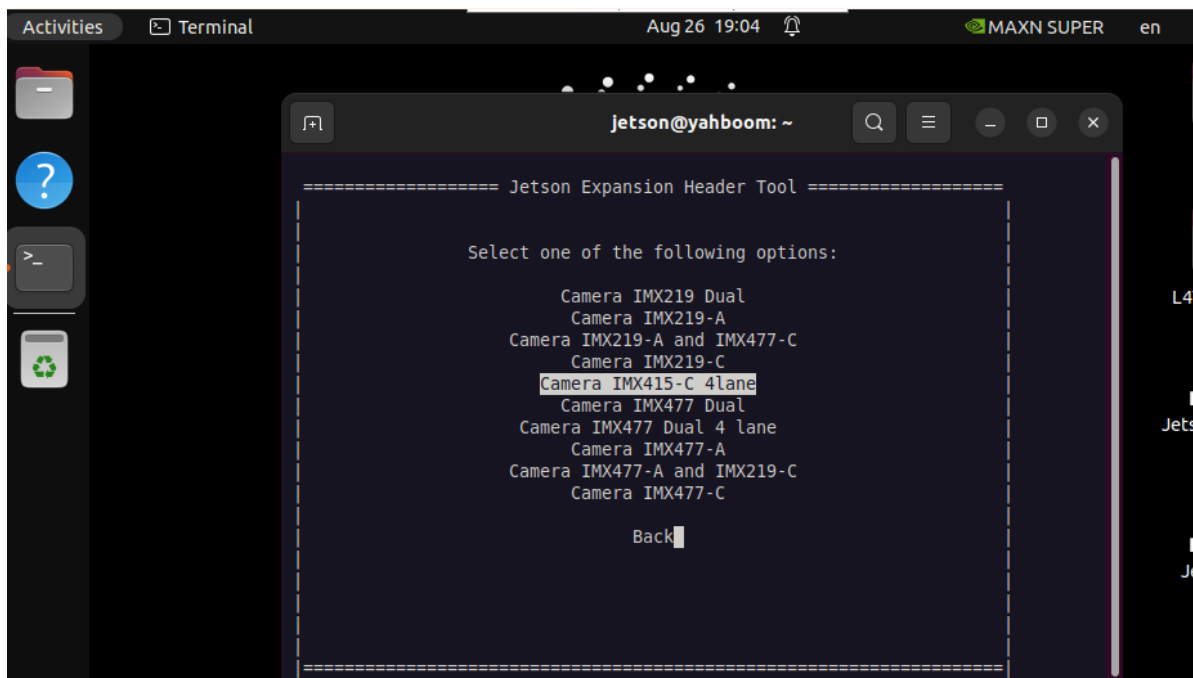
Use the arrow keys to select `Configure Jetson 24pin CSI Connector`, then press `Enter` to enter configuration:



Select `Configure for compatible hardware`, then press `Enter` to enter configuration:



Select `camera IMX219 Dva1`, then press `Enter` to enter configuration:



Select `save pin changes`, then press `Enter` to enter configuration:



Select `save and reboot to reconfigure pins`, then wait for the system to reboot:



## 8. Reboot

```
sudo reboot
```

If `JetsonIO` appears in the boot menu, select it; if only the default item boots automatically, confirm that `DEFAULT JetsonIO` is in effect.

## 9. Check Whether the Driver Is Loaded

After reboot, run:

```
lsmod | grep -E 'nv_imx415|tegra_camera'
```

Expected output is similar to:

```
nv_imx415                20480  0
tegra_camera             217088  4
nv_imx415,nvhost_isp5,nvhost_nvcsi_t194,nvhost_vi5
```

The `0` after `nv_imx415` indicates that there is currently no camera stream, which is normal. The `used by` value only changes after a video stream is opened.

View module information:

```
modinfo nv_imx415 | grep -E 'filename|vermagic|depends'
```

Should show:

```
filename: /lib/modules/5.15.148-tegra/updates/drivers/media/i2c/nv_imx415.ko
depends: tegra-camera
vermagic: 5.15.148-tegra SMP preempt mod_unload modversions aarch64
```

## 10. View Kernel Logs

```
dmesg | grep -Ei 'imx415|nvcsi|vi5|isp5|tegra-camera'
```

If the following appears:

```
dmesg: read kernel buffer failed: operation not permitted
```

It means the current user has no direct permission to read dmesg. Use `sudo`:

```
sudo dmesg | grep -Ei 'imx415|nvcsi|vi5|isp5|tegra-camera'
```

If `sudo dmesg` still fails, you can temporarily disable the dmesg restriction:

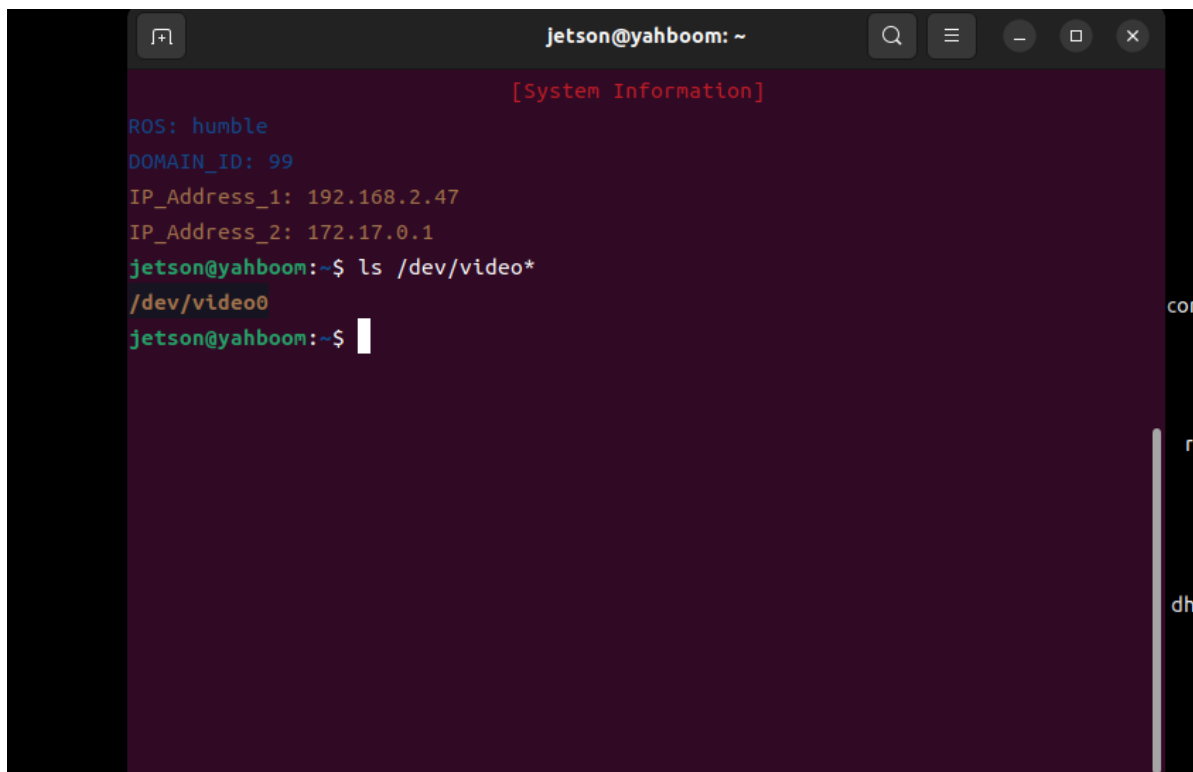
```
sudo sysctl kernel.dmesg_restrict=0
sudo dmesg | grep -Ei 'imx415|nvcsi|vi5|isp5|tegra-camera'
```

You can also view the kernel logs:

```
journalctl -k -b | grep -Ei 'imx415|nvcsi|vi5|isp5|tegra-camera'
```

## 11. Check the Camera Device Node

```
ls /dev/video*
```

A screenshot of a terminal window titled 'jetson@yahboom: ~'. The terminal shows system information including ROS version (humble), domain ID (99), and IP addresses. The user then runs the command 'ls /dev/video\*' which outputs '/dev/video0'.

```
jetson@yahboom: ~  
[System Information]  
ROS: humble  
DOMAIN_ID: 99  
IP_Address_1: 192.168.2.47  
IP_Address_2: 172.17.0.1  
jetson@yahboom:~$ ls /dev/video*  
/dev/video0  
jetson@yahboom:~$
```

## 12. Test the Image Stream

First find the device number corresponding to the IMX415, then test using the following method:

```
nvgstcapture-1.0
```

```
Unknown Aug 21 12:04 MAXIN SUPER 英
jetson@yahboom: ~
** Message: 12:03:51.274: <main:4732> iterating capture
GST_ARGUS: Creating output stream
CONSUMER: Waiting until producer is connected...
GST_ARGUS: Available Sensor modes :
GST_ARGUS: 3864 x 2192 FR = 59.999999 fps Duration = 160
e min 0.000000, max 72.000000; Exposure Range min 15000

GST_ARGUS: 3864 x 2192 FR = 59.999999 fps Duration = 160
e min 0.000000, max 72.000000; Exposure Range min 15000

GST_ARGUS: Running with following settings:
  Camera index = 0
  Camera mode  = 1
  Output Stream W = 3864 H = 2192
  seconds to Run   = 0
  Frame Rate = 59.999999
GST_ARGUS: Setup Complete, Starting captures for 0 seconds
GST_ARGUS: Starting repeat capture requests.
CONSUMER: Producer has connected; continuing.
█
```

## 13. FAQ

### 13.1 `nv_imx415` Is Not Loaded

Load it manually:

```
sudo modprobe nv_imx415
```

If it fails, check:

```
ls -l /lib/modules/5.15.148-tegra/updates/drivers/media/i2c/nv_imx415.ko
sudo depmod -a
modinfo nv_imx415 | grep vermagic
```

If the kernel version is not `5.15.148-tegra`, you need to find an `nv_imx415.ko` that matches that kernel; do not use the precompiled file in the repository directly.

### 13.2 Driver Loaded but No `/dev/video*`

The most common causes are that the overlay did not take effect or the sensor probe failed. Check:

```
cat /boot/extlinux/extlinux.conf
ls -l /boot/tegra234-p3767-camera-p3768-imx415-C-4lane.dtbo
sudo dmesg | grep -Ei 'imx415|tegra-vi|nvcsi|isp5'
```

You can also check whether the imx415 node appears in the device tree:

```
find /proc/device-tree -iname '*imx415*'
```

If dmesg shows an I2C probe failure, first check the ribbon cable, camera power supply, and interface direction.

## 13.3 dmesg Permission Issue

Use:

```
sudo dmesg
```

If it still reports `Operation not permitted`:

```
sudo sysctl kernel.dmesg_restrict=0
```

This setting is restored after reboot and is only used for temporary log viewing.

## 13.4 `nv_imx415` Shows `Used by: 0`

This is normal. `0` means the module is loaded but no application is currently opening a camera stream; it does not mean the sensor is not recognized. Run `v4l2-ctl` or a GStreamer capture once and check again.

## 13.5 Overlay Not Loaded After Boot

Confirm that neither the `DEFAULT JetsonIO` line nor the `OVERLAYS` line in `/boot/extlinux/extlinux.conf` is commented out, and confirm that the corresponding dtbo filename and path match exactly. A reboot is required after modification.

## 13.6 Switch Back to the Original Configuration

Restore the backup:

```
sudo cp /boot/extlinux/extlinux.conf.bak /boot/extlinux/extlinux.conf
sudo reboot
```

## 14. Notes

- The dtbo in the repository is prepared for `p3767/p3768`, HDR40 configuration, and 4-lane IMX415. Do not use it directly on other carrier boards.
- `nv_imx415.ko` depends on `tegra-camera`, so `tegra_camera` must be loaded.
- Always back up before modifying extlinux to avoid the Orin failing to boot normally.